

RMA TRACKER DOCUMENTATION LIBRARY

VERSION 2.1.1

VOLUME III

User Manual and Administration Guide



RMA Tracker

Provides a comprehensive reference for the operation, administration, reporting, backup, troubleshooting, and day-to-day use of RMA Tracker Version 2.1.1.

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1. Introduction

The RMA Tracker Version 2.1.1 User Manual & Administration Guide provides comprehensive instructions for operating, administering, and maintaining the RMA Tracker desktop application. It serves as the primary operational reference for users responsible for managing Return Merchandise Authorization (RMA) records throughout their lifecycle.

RMA Tracker was developed by Brennen Gabriel for Harp Enterprises to provide a centralized solution for managing repair requests, equipment tracking, shipping information, reporting, and database maintenance. This manual explains application features, recommended workflows, reporting capabilities, backup procedures, troubleshooting techniques, and administrative responsibilities.

1.1. Purpose

The purpose of this manual is to provide complete operational guidance for RMA Tracker Version 2.1.1. It explains day-to-day operation, reporting, exports, backups, restoration, troubleshooting, and administrative maintenance.

1.2. Scope

This manual covers application startup, navigation, RMA management, searching, reporting, exporting, backup and restore, troubleshooting, and operational best practices. Technical implementation is covered in Volume II.

1.3. Intended Audience

This manual is intended for end users, repair technicians, supervisors, managers, system administrators, and trainers.

1.4. Relationship to the Documentation Library

Volume I defines software requirements; Volume II documents technical design; Volume III explains operation and administration; Volume IV covers installation, deployment, and maintenance.

1.5. Document Conventions

Bold text identifies interface elements. Italics identify document titles or new terms. Monospaced text identifies filenames and paths. Notes, warnings, and tips provide additional guidance.

1.6. Definitions, Acronyms, and Abbreviations

RMA – Return Merchandise Authorization

SQLite – Embedded relational database

PDF – Portable Document Format

Excel – Microsoft Excel workbook (.xlsx)

1.7. References

- Volume I – Software Requirements Specification
- Volume II – Technical Design & Architecture Manual
- Java Documentation
- SQLite Documentation
- Gradle User Manual

2. Getting Started

This section introduces new users to RMA Tracker Version 2.1.1 and explains how to prepare, launch, navigate, and begin using the application. It describes the main user interface, system requirements, data storage, and recommended first-use procedures so users can begin managing Return Merchandise Authorization (RMA) records confidently and consistently.

2.1. Application Overview

RMA Tracker is a Java desktop application developed for Harp Enterprises to manage the complete lifecycle of Return Merchandise Authorizations (RMAs). It centralizes customer, repair, shipping, status history, reporting, and backup information in a single SQLite database. The application is designed for reliability, ease of use, and fast access to repair records.

2.2. System Requirements

Minimum Requirements:

- Windows 10/11, macOS, or supported Linux distribution
- Java Runtime (if not bundled)
- 4 GB RAM (8 GB recommended)
- 500 MB free disk space
- 1280×720 display or higher

Recommended: SSD storage, 1920×1080 display, daily database backups.

2.3. Starting the Application

Locate the installed RMA Tracker application or packaged executable and launch it. During startup the application initializes configuration files, opens the SQLite database, verifies required folders, and prepares the main window. If automatic backups are enabled, the application also validates the backup environment.

2.4. Main Window Overview

The main window provides access to all primary features. Users can view the RMA list, search records, filter by status, create or edit RMAs, open reports, export data, and access backup and restore functions.

2.5. Navigation

Navigation is designed around a single primary window. Double-click an RMA to edit it, use toolbar buttons and menus for common actions, and return to the main table after closing dialogs.

2.6. Understanding the Dashboard

The dashboard displays current RMA records in a sortable table. Selecting a record displays its associated repair items and provides quick access to editing, reporting, and exports.

2.7. Menus and Toolbar

The menu bar groups application commands into logical categories such as File, Reports, Backup, and Help. Frequently used actions are also available from toolbar buttons and context menus.

2.8. First-Time Setup

On first launch, verify that the database location, backup folder, and application settings are correct. If migrating from an earlier version, restore the latest verified backup before entering new data.

2.9. Data Storage Overview

Application data is stored locally in an SQLite database. Reports, exports, and backup files are stored in dedicated folders created automatically by the application.

2.10. First Use Checklist

- Launch the application successfully.
- Verify database connectivity.
- Confirm backup location.
- Create a sample RMA.
- Add a repair item.
- Generate a report.
- Export to PDF and Excel.
- Perform a manual backup.
- Confirm automatic backups are functioning.

3. Creating and Managing RMAs

This section explains how to create, manage, and maintain Return Merchandise Authorization (RMA) records in RMA Tracker Version 2.1.1. It describes the complete workflow from creating a new RMA through recording repair items, shipping information, validation, and recommended operational best practices.

3.1. RMA Lifecycle

The lifecycle of an RMA begins when equipment is received for service. Users create an RMA, add repair items, update shipping information, track status changes, record repairs, generate reports, and close the record once all work has been completed and returned to the customer.

3.2. Creating a New RMA

To create a new RMA:

1. Select New RMA from the toolbar or File menu.
2. Enter the RMA number.
3. Enter the date sent and date received.
4. Select the current status.
5. Add shipping information.
6. Add one or more repair items.
7. Save the record.

After saving, the RMA becomes searchable and available for reporting.

3.3. General Information

Each RMA includes identifying information such as the RMA number, dates, status, notes, and tracking numbers. These fields should be completed accurately because they are used by the search system and reports.

3.4. Shipping Information

Shipping information records outgoing and return carrier tracking numbers. Verify tracking numbers before saving to reduce shipping errors and improve traceability.

3.5. Adding Repair Items

Each repair item represents a single machine or device. Record the county or customer, serial number, machine type, software version (if applicable), problem description, repair description, and whether the item has been received.

3.6. Machine Types

Select the appropriate machine type from the predefined list to ensure consistency across reports and searches. Use standardized terminology when entering equipment information.

3.7. Saving an RMA

Before saving, verify all required fields have been completed. Selecting Save writes the RMA and its repair items to the SQLite database. Any validation errors must be corrected before the operation completes.

3.8. Validation Messages

RMA Tracker validates required fields and displays descriptive error messages if information is missing or invalid. Users should review these messages carefully before attempting to save again.

3.9. Best Practices

- Use a consistent RMA numbering convention.
- Enter complete shipping information whenever available.
- Record detailed but concise problem and repair descriptions.
- Update the RMA status immediately after workflow changes.
- Verify data before generating reports or exports.
- Perform regular backups to protect operational data.
- Review completed RMAs before closing them to ensure accuracy.

4. Viewing and Editing RMAs

This section explains how to locate, open, review, edit, and maintain existing Return Merchandise Authorization (RMA) records. It covers updating general information, repair items, shipping details, status history, and saving changes while preserving data integrity.

4.1. Opening Existing RMAs

Existing RMAs can be located from the main application window using the search field, status filter, or the main RMA table. Select the desired record and either double-click it or choose the Edit command to open the RMA editor.

4.2. Double-Click Navigation

Double-clicking an RMA record opens the Edit RMA window, allowing users to review and modify the selected record without navigating through additional menus. This provides the quickest method for accessing existing RMAs.

4.3. Editing Information

Users may update the RMA number (when permitted), dates, notes, status, tracking numbers, and repair information. After making changes, review the updated information for accuracy before saving.

4.4. Updating Shipping

Shipping information may be updated throughout the repair lifecycle. Verify carrier names and tracking numbers before saving. Maintain both outgoing and return shipment information whenever available to preserve a complete audit trail.

4.5. Managing Repair Items

Repair items can be added, modified, or removed as work progresses. Update machine type, serial number, problem description, repair description, and the Received indicator as required. Ensure each repair item accurately reflects the work performed.

4.6. Viewing Status History

Status History displays a chronological record of status changes for the selected RMA. Reviewing the history helps users monitor repair progress, verify workflow completion, and investigate previous updates.

4.7. Saving Changes

Select Save after completing edits. The application validates required fields before committing changes to the SQLite database. If validation fails, correct the highlighted information and save again.

4.8. Canceling Changes

Selecting Cancel closes the edit window without saving any unsaved modifications. Users should confirm they intend to discard changes before exiting.

4.9. Deleting RMAs

Deleting an RMA permanently removes the record and its associated repair information from the active database. This action should be limited to duplicate or incorrectly created records. Review the confirmation prompt carefully before proceeding.

4.10. Data Integrity and Best Practices

Recommended practices include:

- Verify the correct RMA is selected before editing.
- Keep repair descriptions concise but complete.
- Update the RMA status immediately after workflow changes.
- Verify tracking numbers before saving.
- Review Status History after significant updates.
- Generate a report to verify important edits.
- Perform regular backups before making

5. Searching and Filtering

This section explains how to locate RMAs quickly using the application's search and filtering capabilities. Effective searching allows users to retrieve records by RMA number, county, serial number, machine type, tracking number, status, and descriptive text while reducing the time required to locate historical information.

5.1. Search Overview

The search system performs searches across RMA information and associated repair items. Users can combine searches with status filters to narrow results.

5.2. Search by RMA Number

Enter the RMA number in the search field to locate a specific record.

5.3. Search by County

Search using the county or customer name to locate all associated repair items and RMAs.

5.4. Search by Serial Number

Enter a complete or partial serial number to locate an individual machine.

5.5. Search by Machine Type

Filter records by equipment type to identify similar devices.

5.6. Search by Tracking Number

Locate RMAs using outgoing or return shipment tracking numbers.

5.7. Search by Problem or Repair Description

Search descriptive text to locate repairs involving specific issues or completed work.

5.8. Status Filtering

Use the Status filter to display Ready to Ship, Shipped, In Repair, Repair Complete, Shipped Back, Received Back, Closed, or all records.

5.9. Combined Searches

Combine keyword searches with status filters for precise results.

5.10. Clearing Search Results

Clear the search field and reset the status filter to display all records.

5.11. Best Practices

Use specific search terms, verify spelling, keep descriptions consistent, and avoid unnecessary abbreviations to improve search accuracy.

6. Status and Workflow Management

This section documents the RMA lifecycle and explains how status values are used to monitor progress from receipt through repair and final closure. Maintaining accurate status information provides visibility into workload, improves reporting, and ensures consistent operational procedures.

6.1. Workflow Overview

Each RMA progresses through a defined workflow. Users should update the status whenever a significant stage of the repair process is completed.

6.2. Ready to Ship

The RMA has been prepared for shipment but has not yet left the facility.

6.3. Shipped

The equipment has been shipped to the repair facility or destination.

6.4. In Repair

Equipment has been received and repair work is actively underway.

6.5. Repair Complete

All required repair work has been completed and the equipment is awaiting return shipment.

6.6. Shipped Back

The repaired equipment has been shipped back to the customer.

6.7. Received Back

Confirmation has been received that the repaired equipment has arrived at its destination.

6.8. Closed

The RMA has completed its lifecycle and requires no additional action.

6.9. Status History

Every status update is recorded chronologically, allowing users to review the complete history of an RMA.

6.10. Recommended Workflow

Create RMA → Ready to Ship → Shipped → In Repair → Repair Complete → Shipped Back → Received Back → Closed.

6.11. Best Practices

Update statuses immediately after workflow changes, avoid skipping stages when possible, verify shipping before marking records as shipped, and review status history before closing an RMA.

7. Reports, Printing, and Print Preview

This section explains how to generate reports, review them using Print Preview, configure print settings, and produce professional printed documentation from RMA Tracker. Reports provide a complete summary of RMA information, repair items, shipping details, and status history for operational and administrative use.

7.1. Reporting Overview

RMA Tracker generates reports for individual RMAs and current search results. Reports present information in a consistent, printable format suitable for record keeping and customer communication.

7.2. Viewing Reports

Select an RMA and choose View Report or Report Preview. The report is generated using the current database information and displayed before printing.

7.3. Report Contents

Reports may include the RMA number, dates, status, customer or county information, repair items, machine types, serial numbers, problem descriptions, repair descriptions, shipping information, notes, and status history.

7.4. Print Preview

Print Preview allows users to review formatting before printing. Users can verify page breaks, headers, footers, margins, and overall layout.

7.5. Page Setup

Use Page Setup to configure paper size, page orientation, and margins. Verify settings before printing to reduce paper waste.

7.6. Printing Reports

Select Print from the toolbar or preview window, choose a printer, configure print options, and confirm the output. Review the first printed page before printing multiple copies.

7.7. Printing Current Results

The current filtered RMA list can be printed to document active work or create management reports.

7.8. Troubleshooting Printing

If printing fails, verify printer connectivity, paper availability, printer selection, and permissions. Use Print Preview to confirm report formatting before retrying.

7.9. Best Practices

Review reports before printing, use Print Preview for every report, maintain consistent page settings, and archive electronic copies when appropriate.

8. Exporting Data

This section describes how to export RMA information to PDF and Microsoft Excel. Exports allow users to share information, archive records, and perform additional analysis outside the application while preserving important operational data.

8.1. Export Overview

RMA Tracker supports exporting individual RMAs and current search results. Exported files provide portable copies of operational data.

8.2. Exporting to PDF

Choose Export to PDF, select the destination folder, confirm the filename, and save the report. PDF exports preserve report formatting for sharing and printing.

8.3. Exporting to Microsoft Excel

Choose Export to Excel to generate an XLSX workbook. Excel exports support additional filtering, sorting, and analysis.

8.4. Export Locations

Exports are saved to the user-selected folder or the application's default export directory. Verify write permissions before exporting.

8.5. File Naming

Use meaningful filenames that include the RMA number or report purpose and, when appropriate, the export date to simplify retrieval.

8.6. Verifying Exported Files

Open the exported PDF or Excel file to confirm that all expected information has been included and formatted correctly.

8.7. Common Troubleshooting

If an export fails, verify disk space, folder permissions, and that the destination file is not currently open in another application.

8.8. Best Practices

Store exports in organized folders, avoid overwriting important files, verify exported information before distribution, and protect files containing sensitive business information.

9. Backup and Restore

This section explains how to protect application data using the backup and restore features. Regular backups help prevent data loss and allow recovery after hardware failures, accidental deletion, or software issues.

9.1. Backup Overview

RMA Tracker stores operational data in an SQLite database. The application supports both automatic and manual backups to protect this database.

9.2. Automatic Backups

Automatic backups are created during normal application operation according to the configured behavior. Verify that automatic backups complete successfully and are stored in the expected location.

9.3. Manual Backups

Select the Backup command from the application to create an immediate backup before upgrades or major data changes. Confirm the backup completes successfully.

9.4. Backup File Locations

Backups are stored in the application's backup directory. Administrators should periodically verify the folder exists, has sufficient free space, and is included in organization-wide backup policies.

9.5. Restore Process

Choose Restore, browse to the desired backup, confirm the selected file, and allow the application to replace the active database. Restart the application if prompted.

9.6. Backup Verification

After creating a backup, verify the file exists, the timestamp is correct, and the backup can be opened or restored in a test environment when appropriate.

9.7. Recovery Best Practices

Keep multiple backup generations, store copies on separate media when possible, test restores periodically and always create a backup before software upgrades.

10.Application Administration

This section describes routine administrative responsibilities required to keep RMA Tracker operating reliably and securely.

10.1. Administrator Responsibilities

Administrators maintain the application, verify backups, assist users, monitor data integrity, and coordinate upgrades.

10.2. Database Management

Monitor database size and integrity, ensure the database remains accessible, and restore from verified backups if corruption occurs.

10.3. File Locations

Understand the locations of the application database, backup folders, exported reports, and generated PDF/Excel files.

10.4. Routine Maintenance

Review backup logs, remove obsolete exports if required, install approved software updates, and verify application functionality after maintenance.

10.5. Upgrading

Before upgrading, create a verified backup, close the application, install the new version, confirm the database opens successfully, and validate critical workflows.

10.6. Administrative Best Practices

Limit administrative access, maintain version records, document maintenance activities, and periodically review storage capacity.

11.Troubleshooting

This section provides guidance for diagnosing and resolving common issues encountered while using RMA Tracker.

11.1. Startup Problems

If the application fails to start, verify the installation, confirm required files are present, and review any displayed error messages.

11.2. Database Problems

If the database cannot be opened, confirm the file exists, verify permissions, and restore the most recent verified backup if necessary.

11.3. Search and Reporting Problems

Ensure search criteria are correct, verify records exist, and confirm reports are generated from the current dataset.

11.4. Export Problems

If PDF or Excel exports fail, verify the destination folder is writable and no exported file is currently open in another application.

11.5. Backup and Restore Problems

Confirm sufficient disk space, verify the selected backup file is valid, and avoid interrupting backup or restore operations.

11.6. Performance Issues

Close unnecessary applications, verify sufficient system resources, and archive or optimize data if exceptionally large datasets affect performance.

11.7. Contacting Support

When reporting a problem, include the application version, operating system, steps to reproduce the issue, screenshots when available, and any displayed error messages.

12. Frequently Asked Questions (FAQ)

This section answers common questions encountered by users and administrators while operating RMA Tracker.

12.1. General

Q: What is an RMA?

A: A Return Merchandise Authorization record used to track equipment throughout the repair lifecycle.

Q: Can multiple repair items belong to one RMA?

A: Yes. An RMA can contain multiple machines or repair items.

12.2. RMAs

Q: Can I edit an RMA after saving?

A: Yes, open the record from the main window and save your changes.

Q: Can I delete an RMA?

A: Yes, but only when appropriate because deletion permanently removes the record.

12.3. Reports

Q: Why doesn't my report include recent changes?

A: Save the RMA before generating the report.

Q: Can I print filtered results?

A: Yes, current search results can be printed.

12.4. Backup

Q: How often should backups be performed?

A: Automatic backups should remain enabled and manual backups should be created before upgrades or major changes.

Q: How do I restore data?

A: Use the Restore function and select a verified backup.

12.5. Administration

Q: Where is the database stored?

A: In the application's data directory.

Q: Who should perform upgrades?

A: A system administrator.

13. Best Practices

Following these recommendations helps maintain accurate data, improve workflow consistency, and reduce the likelihood of operational errors.

13.1. Data Entry

Use consistent naming conventions, complete all required fields, and verify information before saving.

13.2. Naming Standards

Maintain consistent RMA numbering and descriptive repair notes.

13.3. Status Management

Update status immediately after workflow changes and review status history regularly.

13.4. Backup Schedule

Allow automatic backups to run and perform manual backups before upgrades or bulk

13.5. Data Quality

Review completed RMAs for accuracy, verify tracking numbers, avoid duplicate records, and archive exported reports appropriately.

14. Appendices

The appendices provide supplemental reference material supporting the use and administration of RMA Tracker.

14.1. Keyboard Shortcuts

Document all supported keyboard shortcuts and menu accelerators.

14.2. Status Reference

Summary table describing each RMA status and its purpose within the workflow.

14.3. Machine Type Reference

Reference list of supported machine types used throughout the application.

14.4. Error Messages

Catalog of common validation and runtime messages with recommended corrective actions.

14.5. Glossary

Definitions of technical terms, abbreviations, and application-specific terminology.

14.6. Screenshot Catalog

Annotated screenshots of the Main Window, Create/Edit RMA dialog, Report Preview, Backup & Restore, and Export windows.

14.7. Sample Workflow

Example showing the complete lifecycle of an RMA from creation through closure.

14.8. Revision History

Summary of document revisions, approval dates, and version changes.